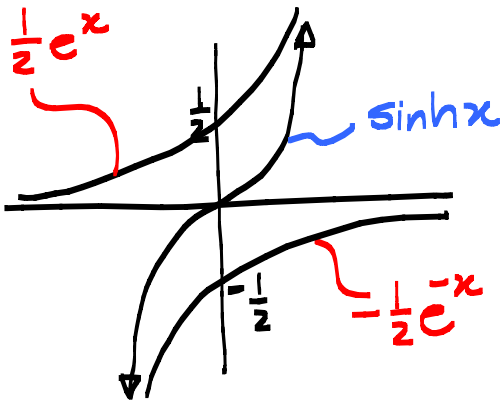


Unit 3.7 Hyperbolic functions p 254

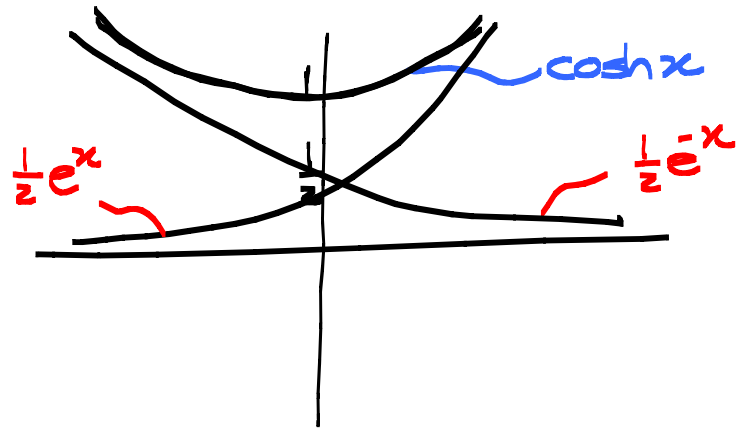
Note Title

2015/04/07

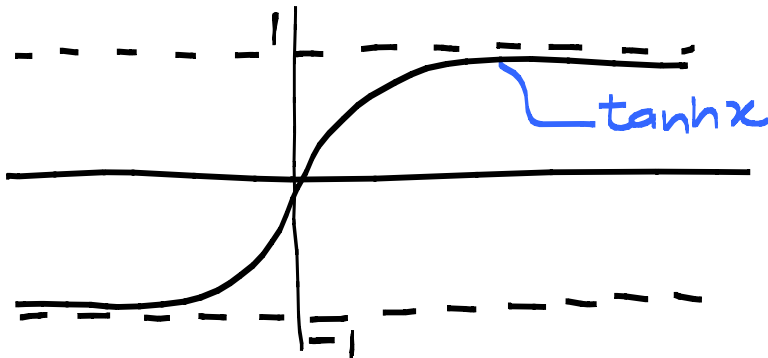
$$\sinh x := \frac{e^x - e^{-x}}{2}$$



$$\cosh x := \frac{e^x + e^{-x}}{2}$$



$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Derivatives [Proof - SELF]

$$y = \sinh x$$

$$\frac{dy}{dx} = \cosh x$$

$$y = \cosh x$$

$$\frac{dy}{dx} = \sinh x$$

$$y = \tanh x$$

$$\frac{dy}{dx} = \operatorname{sech}^2 x$$

Examples:

$$\textcircled{1} \quad y = \tanh^{-1}(\sinh x) \Rightarrow \frac{dy}{dx} = \frac{\cosh x}{1 + (\sinh x)^2}$$

② $y = \cosh(\sin^{-1}x)$

$$\frac{dy}{dx} = \sinh(\sin^{-1}x) \cdot \frac{1}{\sqrt{1-x^2}}$$

